

Lab 08

Due March 24, 5:00 PM PST

AUTHOR

Peter Khrissanov 301611434

```
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

```
✓ dplyr      1.2.0      ✓ readr      2.1.6
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.2      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.1
```

— Conflicts — tidyverse_conflicts() —

```
* dplyr::filter() masks stats::filter()
```

```
* dplyr::lag() masks stats::lag()
```

i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
library(tidytext)
library(janeaustenr)
book_words <- austen_books() |>
unnest_tokens(word, text)
```

Problem 1a

```
SandS <- book_words |> filter(book == "Sense & Sensibility")
allword <- dim(unique(SandS))[1]
Tenchar <- SandS |> filter(str_length(word)>= 10)
bigword <- dim(unique(Tenchar[2]))[1]
proportion <- (bigword/allword) * 100
proportion
```

```
[1] 20.43932
```

Problem 1b

```
## write answer here

book_list <- unique(book_words$book)
proportion_list <- vector("double", length(book_list))

for(i in seq_along(book_list)) {
```

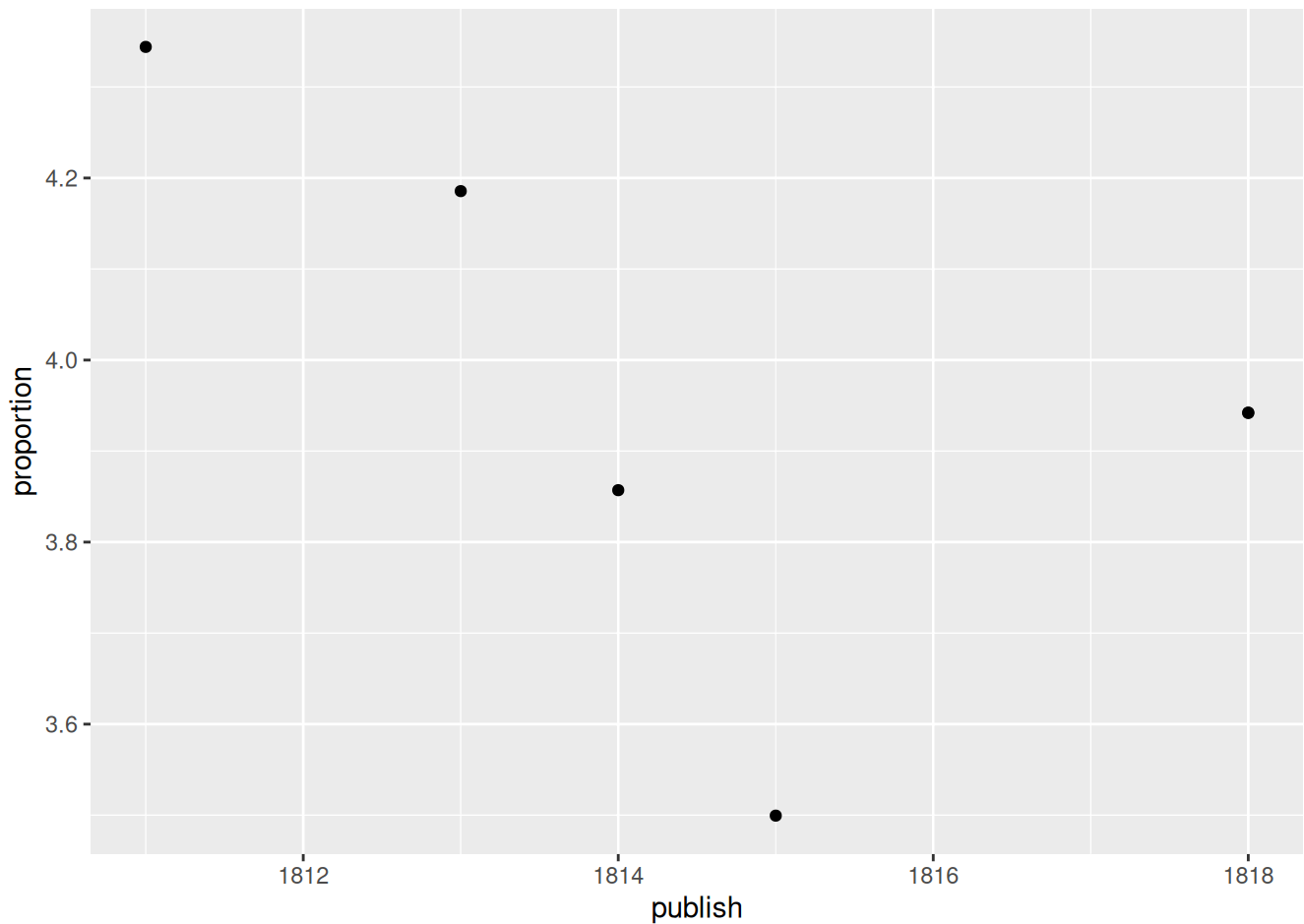
```
one_book <- book_words |> filter(book == book_list[i])

allword <- dim(one_book)[1]
bigword <- one_book |> filter(str_length(word) >= 10)
bigwordnum <- dim(bigword)[1]

proportion_list[i] <- (bigwordnum / allword) * 100
}

plot_df <- data.frame(
  publish = c(1811, 1813, 1814, 1815, 1818, 1818),
  proportion = proportion_list
)

ggplot(plot_df, aes(x = publish, y = proportion)) +
  geom_point()
```



Problem 1c

```
## write answer here
```

```
book_words |>  
  filter(str_detect(word, "[wxyz]{5}e$")) |>  
  distinct(word)
```

```
# A tibble: 4 × 1
```

```
  word
```

```
  <chr>
```

```
1 welfare
```

```
2 welcome
```

```
3 wimpole
```

```
4 warfare
```